PHYS400
SPECIAL PROBLEMS IN PHYSICS
INTERIM REPORT

Project Title	A Pure Analytical Gaussian Expansion Method for Heavy and Mixed-State Quarkonia
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Abstract

Inspection of heavy quarkonia is a great benchmark for the non-perturbative low-energy regime of the Quantum Chromodynamics (QCD). For this project we have been trying to explain heavy mesons and baryons using Cornell potential, a phenomenological model, that has no purely analytical solution due to Coulomb-plus-linear structure. Through this project, we have seen that Rayleigh-Ritz variational method can be applied to this potential very well; however, imposing Gaussian Expansion Method showed much more fruitful results. Furthermore, we have used a mixed method of perturbative analysis with variational analysis. $O(\alpha_s^2)$ relativistic corrections (Breit-Fermi interaction) were analytically integrated where possible.

The code that has been generated (`gem_solver.py`, `fitter.py`, `observables.py`) is working exceptionally well for this project. Through χ^2 optimization, we achieved to fit our Cornell potential parameters to 2024 Particle Data Group (PDG) tables. The found wavefunction expansions exceptionally showed sub-1% error for all existing states. Also, even if there is no experimental data, the code generates all 30-state for even unobserved S, P, and D excitations at once. Also, with the usage of Hypervirial Theorem, we have showed that our selected parameters for the Hamiltonian works in well-adjusted limits.

The ongoing search for the explanation of the heavy quarkonia will continue with the calculation of the decay widths, root-mean-square (RMS) radii and extending the project to a three-body problem, solving for the heavy baryonic states, with the usage of the Jacobi Coordinates.

Introduction

Internal dynamics of the hadronic states such as, as in our problem, heavy meson and baryons are studied under QCD, a non-Abelian gauge thery that includes asymptotic freedom at high energies and color confinement on large scales. In the non-relativistic, i.e low-energy, regime the strong coupling constant, α_s, evidently is not constant but even diverges. This divergence results on failure of the perturbative solutions and also analytical solutions may become problematic in the sense that they cannot be solvable. This results the need for non-perturbative methods such as lattice QCD simulations, effective field theories, or phenomenological potential models such as, in our case, Cornell potential (Elpe, 2016; Bali, 2001).

Heavy mesons, especially the bottomonium and charmonium sector that contains $b\bar{b}$ and $c\bar{c}$ excitations, and the mixed state, provide us the perfect test ground for such phenomenological models. Because of their constituent masses ($m_b \approx 4.73$ GeV, $m_c \approx 1.50$ GeV) exceeds QCD confinement scale ($\Lambda_{QCD} \approx 200$ MeV), we can treat these sectors in a non-relaticistic sense (Griffiths, 1987; Quigg and Rosner, 1979)

As mentioned, we will use vastly successful Cornell potential model, that includes short-range Coulomb-like interaction derived from one-gluon exchange and long-range linear term that implements color confinement (Eichten et al., 2008). While there is already vast literature on this potential, variational method applications, supersymmetric analytic solutions (Vega and Flores, 2014; Akbar et al., 2025), the usage of the numerical integration methods fail due to $1/r$ and α_s divergences near $r \to 0$. Thus we require a somewhat analytical and somewhat numerical approaches as we will be doing while imposing Gaussian Expansion Method (GEM) and its possible integral solutions vastly.

Research Question and Aims:

We started this project to benchmark Rayleigh-Ritz variational method over heavy meson systems; however, also imposing the GEM, we have observed that we do not need to find ansätz functions one by one but can generate them using Gaussian bases. Solving this problem was still numerically impossible so we have managed to impose analytical solutions to the mass spectra by using Gaussian integrals that arises from inner product of $\langle\psi|H\psi\rangle$. This method showed to be much stronger than other numerical methods that utilizes perturbative or one-term ansätz functions. Thus we can create a new question: Can we create a generalized few-body quarkonia setup solver with GEM? If we can what would be the benefits and computational and analytical limitations? On what limit should we consider GEM useful? Can we use this setup for other problems, e.g. quark-gluon plasma? In this semester we are planning to propose a solution for heavy meson, baryon, and if possible tetraquark setups.

Theoretical Framework

To model our heavy meson problem, we will adopt a non-relativistic approach using the Cornell potential. This potential can be seen from Equation 1, where p is the combined state momentum, $\mu = \frac{m_1 m_2}{m_1+m_2}$ is the reduced mass, b is the QCD string tension and c is the simple potential shift.

$$H_0 = \frac{p^2}{2\mu} - \frac{4}{3}\frac{\alpha_s}{r} + \sigma r + c \tag{1}$$

This potential includes the single color Coulombic interaction and the linear confinement by QCD string tension very well (Eichten et al., 1975; Alkathiri et al., 2024).

The Gaussian Expansion Method (GEM)

To solve the stationary Schrödinger equation $H_0\psi = E\psi$, the anätz wave functions are chosen as some non-orthogonal linear combination of Gaussian basis vectors as can be seen from Equation 2.

$$\psi_{lm}(\mathbf{r}) = Y_{lm}(\hat{r}) \sum_{n=1}^{N} c_n \, r^l e^{-\nu_n r^2} \tag{2}$$

To enforce this basis to *almost* complete and capable of analytical problems such as the divergence at $r \to 0$, the width parameter ν_n are chosen in geometric progression $\nu_n = 1/(r_{min}^2 a^{n-1})$. This also prevents ill-conditioned overlap of the chosen basis functions and also governs both Coulombic divergence scale and the linearly confined scale (Hiyama et al., 2003). Also, as mentioned, GEM method has been proved useful in nuclear and atomic physics, especially non-relativistic regime like hypernuclei, e.g ^{4}He, setups (Hiyama, 2012). This makes the method promising for many-quark hadronic states.

Analytical Matrix Elements & Elimination of Coulomb Singularities

Through our process until now, we have observed that numerical integration of the Coulombic contribution results in singularity, even if not it creates ill-conditioned generalized eigenvalue problems that make the Gaussian bases non-complete. Attempting the calculate $\langle\psi| - 1/r|\psi\rangle$ using Simpson's rule or Runge-Kutta results in major problems that cannot be fixed by simple solutions, especially we progress towards the origin.

To adress this problem, we have managed to write down a general integral, Equation 3 that can be pure analytically computed for kinetic, T_{ij}, Coulombic, V_c, and confinement, V_l, and fine correction matrices to a zero floating point or truncation errors.

$$\mathcal{I}_p(\nu_{ij}) = \int_0^\infty r^p e^{-\nu_{ij} r^2} dr = \frac{1}{2} \frac{\Gamma\left(\frac{p+1}{2}\right)}{\nu_{ij}^{(p+1)/2}} \tag{3}$$

where $\nu_{ij} = \nu_i + \nu_j$ combined width of the matrix element.

Relativistic Fine and Hyperfine Structure

To adress the problem with only using Cornell potential spin degeneracies, such as 1^3S_1 J/ψ and 1^1S_0 η_c, $O(\alpha_s)$ corrections added first as perturbations but then, except for the hyperfine structure that will be mentioned, analytically (Lucha et al., 1991; Voloshin, 2008). This energy difference shift due to Breit-Fermi interaction can be seen from Equation 4.

$$\Delta E = \langle \psi | V_{SS} + V_{LS} + V_{Tensor} | \psi \rangle \tag{4}$$

Even though we can try to write down this interaction purely analytically, the hyperfine interaction contains a Dirac delta function: $V_{SS} \propto \frac{\alpha_s}{m_1 m_2} \vec{S}_1 \cdot \vec{S}_2 \, \delta^3(\vec{r})$. For obvious reasons we cannot directly evaluate this delta function and we need to use a smeared version of it that is appropriate. This delocalizes and smears our delta function to a Gaussian function. That can be seen from Equation 5.

$$\delta(\mathbf{r}) \to \left(\frac{\sigma_{smear}}{\sqrt{\pi}} \right)^3 e^{-\sigma_{smear}^2 r^2} \tag{5}$$

We will not be carrying all the features of the delta function; however, this approximation provides to be useful and fruitful as can be seen from our results in Appendix B.

Computational Implementation

The computational implementation has been only done using Python and relative libraries one can check from Appendix D. This repository will be actively used throughout this project.

1. Variational Method Validation through Toy Model (`./qho/`): To understand how our method works, we have used a quarticly disturbed quantum harmonic oscillator (QHO). We observed that variational method showed 10^{-2} error comparing to finite differences integration which proves to be much more successful than the perturbative method.

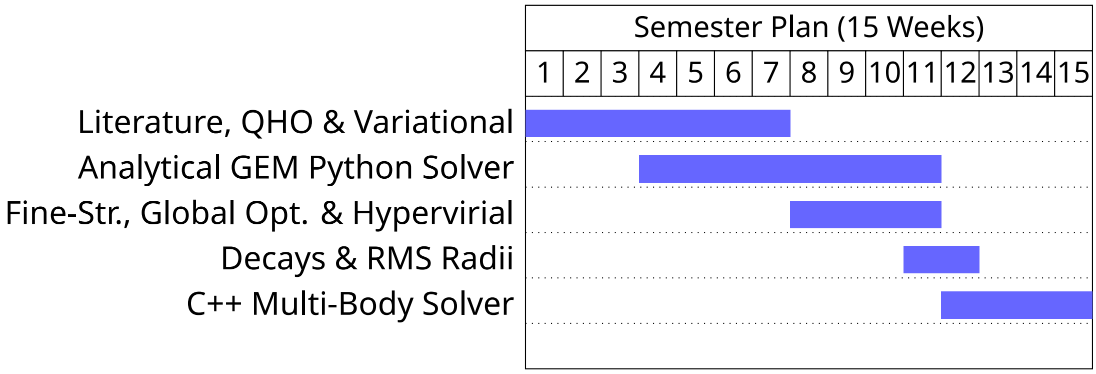
2. Our Modules (`./src/`): This folder contains all our modules. `fitter.py` uses χ^2 optimization to find optimized parameters for the Cornell potential, `gem_solver.py` utilizes GEM method and runs with `observables.py`, which contains our corrections and analytical Gaussian integral, and `metrics.py`, which compares our findings with PDG values, from our main script `./scripts/run_spectrum.py` to find the relevant Gaussian expansion and calculate errors. In general we create a `QuarkoniumSystem` class inside the `gem_solver.py` module and put it through the `gem_solver.py` process. We analytically build H and S matrices for our generalized eigenvalue problem ($Hc = ESc$) using `scipy.special.gamma` and our analytically found integral and solve it using `scipy.linalg.eigh(H,S)`. To prevent ill-conditioned and overlapping results we decided to leave $N = 25$ and also make an exception handling for such cases that if it fails, then we add a small disturbance to the S matrix as $10^{-9}\mathbf{I}$.

3. Observables Engine (`observables.py`): Other than above-mentioned usage, this module includes the wavefunction-at-origin (WFO) as $u'(0) = \sum c_n/N_n$. This evidently removes the need for the grid-based numerical integration methods and allows us to have well-defined states for other usages of our wavefunctions (Ding et al., 1999b).

4. χ^2 Parameter Optimization (`fitter.py` & `metrics.py`): The `fitter.py` module uses the `scipy.optimize.minimize` to operate a χ^2 optimization of the Cornell potential parameters against the PDG masses. The cost/loss function evaluates absolute percentage error. Finally, `metrics.py` formats the results, automatically exporting full spectra in a sector respective CSV file, containing percentage errors (`Pct_Err`) and exact GEM coefficients.

Project Progress

This semester proved to be fruitful for this project. The codebase generated is completely operational for solving meson states. The energy spectra found for all S, P, and D excitations with the χ^2 optimization has been finished well ahead of planned schedule.



Results and Discussion

GEM method resulted in great precision that increased when we transformed all numerical integrations to the simple Gaussian integrals for all of the potential terms. The χ^2 optimization resulted in finding well-defined parameters for Cornell potential which can be seen and mentioned in 6.2. The analytical GEM code yielded phenomenal precision. The global χ^2 optimizer found optimal convergence for Cornell parameters, which have been fully exported (e.g., `bb_params.csv`, `cc_params.csv`) and are completely mapped in Appendix A.

As can be seen from Appendix B, the accuracy and the precision achieved is phenomenal. For the bottomonium sector we have achieved to find $\Upsilon(2S)$ state with 0.0006% error and for the charmonium sector we got J/ψ error as 0.834

The found GEM parameters for 25-basis expansion can be found in Appendix C. As can be observed widths and amplitudes are very well distributed and bounded. This proves that even though we have a cusp, the mathematical stability is wonderful for suggested model.

While we have achieved extremely well energy eigenvalues for the given Hamiltonian, we must also check the validity of our Hamiltonian due to how it has been gathered through parameter optimization. To make this Hamiltonian be a discussion of its own, we have used Hypervirial Theorem (Ding et al., 1999a). The hypervirial theorem resulted in great results that varies on small enough values that we can make this Hamiltonian usable.

Future Outlook

Wavefunction Benchmarking

Even though we have achieved a great deal finding energy expectations close to empirical values, it still is a problem that how much our results are valid. To check this we will simply check decay rates and decay widths, after that checking the lifetime. The reason is that we may have overfitted parameters and this may cause our wavefunctions to work only for the energy expactations.

C++ Multi-Body Porting

In the final part of this research, we will also try to implement three-body problem to our results, e.g Ω_{ccc}^{++}, Λ_c. To mathematically achieve this, we will first impose Jacobi coordinates ($\vec{\rho} = \vec{r}_1 - \vec{r}_2$ and $\vec{\lambda} = \vec{r}_3 - R_{CM_{12}}$) to our problem so that our three-body problem becomes a two-body problem. The GEM is exceptionally worked well with our two-body setup and we are thinking applying this method to understand how heavy baryons work. This method has been used for complex few-body systems such as hypernuclei in the late literature (Hiyama et al., 2003; Hiyama, 2012).

Doing this calculations and derivations even in a three-body system requires computational workload since it increases our problem to a $O(N^3)$ from $O(N)$ or worse due to how our system can or will be structured and additional interaction integrals. While our Python code is highly successful, it has proved to be cumbersome even when doing this two-body problem. Therefore, one milestone of this project will be porting our method to C++, using Eigen library expansively, and possibly creating an API for further use in other programming languages.

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Appendix A: Optimized Cornell Potential Parameters

The `fitter.py` module used `scipy.optimize.minimize` to progress a χ^2 optimization of the Cornell potential parameters against the PDG experimental energy levels. Optimized parameters are recorded inside appropriate CSV files (`bb_params.csv`, `cc_params.csv`, `bc_params.csv`) that are also listed in Table 6.2.

Quarkonium System	α_s	b (GeV^2)	c (GeV)
Bottomonium ($b\bar{b}$)	0.3515	0.2236	-0.0270
Charmonium ($c\bar{c}$)	0.5840	0.1397	-0.0058
Heavy-Light ($b\bar{c}$)	0.4708	0.1592	+0.0336

Table 1: Found Cornell potential parameters for the each sector.

Appendix B: Complete Extracted Mass Spectra

Our GEM solver finds the energy states even if they are not observed or highly unstable. The below calculations are directly done through `run_spectrum.py` using our modules. We have 90 states in total and their absolute and percent errors. You can check the Table 2.

State	Calculated (GeV)	Exp (PDG, GeV)	Abs Error (MeV)	% Error
Charmonium ($c\bar{c}$)				
$(1^1S)\ \eta_c$	2.9841	2.9839	+0.279	0.009%
$(1^3S)\ J/\psi$	3.0710	3.0969	-25.85	0.834%
$(2^1S)\ \eta_c(2S)$	3.6376	3.6375	+0.186	0.005%
$(2^3S)\ \psi(2S)$	3.6662	3.6861	-19.81	0.537%
$(3^1S)\ \eta_c(3S)$	4.0472	N/A	N/A	N/A
$(3^3S)\ \psi(3S)$	4.0660	N/A	N/A	N/A
$(1^1P)\ h_c$	3.5088	3.5254	-16.6	0.470%
$(1^3P_0)\ \chi_{c0}$	3.4177	N/A	N/A	N/A
$(1^3P_1)\ \chi_{c1}$	3.5118	N/A	N/A	N/A
$(1^3P_2)\ \chi_{c2}$	3.5252	3.5562	-31.0	0.871%
$(2^1P)\ h_c(2P)$	3.9779	N/A	N/A	N/A
$(2^3P_0)\ \chi_{c0}(2P)$	3.8893	N/A	N/A	N/A
$(2^3P_1)\ \chi_{c1}(2P)$	3.9784	N/A	N/A	N/A
$(2^3P_2)\ \chi_{c2}(2P)$	3.9954	N/A	N/A	N/A
$(3^1P)\ h_c(3P)$	4.3726	N/A	N/A	N/A
$(3^3P_0)\ \chi_{c0}(3P)$	4.2849	N/A	N/A	N/A

State	Calculated (GeV)	Exp (PDG, GeV)	Abs Error (MeV)	% Error
$(3^3P_1)\ \chi_{c1}(3P)$	4.3717	N/A	N/A	N/A
$(3^3P_2)\ \chi_{c2}(3P)$	4.3907	N/A	N/A	N/A
$(1^1D)\ \eta_{c2}$	3.8076	N/A	N/A	N/A
$(1^3D_1)\ \psi(3770)$	3.8034	N/A	N/A	N/A
$(1^3D_2)\ \psi_2$	3.8151	N/A	N/A	N/A
$(1^3D_3)\ \psi_3$	3.8039	N/A	N/A	N/A
$(2^1D)\ \eta_{c2}(2D)$	4.2180	N/A	N/A	N/A
$(2^3D_1)\ \psi(2D)$	4.2097	N/A	N/A	N/A
$(2^3D_2)\ \psi_2(2D)$	4.2239	N/A	N/A	N/A
$(2^3D_3)\ \psi_3(2D)$	4.2174	N/A	N/A	N/A
$(3^1D)\ \eta_{c2}(3D)$	4.5809	N/A	N/A	N/A
$(3^3D_1)\ \psi(3D)$	4.5699	N/A	N/A	N/A
$(3^3D_2)\ \psi_2(3D)$	4.5857	N/A	N/A	N/A
$(3^3D_3)\ \psi_3(3D)$	4.5822	N/A	N/A	N/A
Bottomonium ($b\bar{b}$)				
$(1^1S)\ \eta_b$	9.3997	9.3987	+1.060	0.011%
$(1^3S)\ \Upsilon(1S)$	9.4291	9.4603	-31.13	0.329%
$(2^1S)\ \eta_b(2S)$	10.0137	N/A	N/A	N/A
$(2^3S)\ \Upsilon(2S)$	10.0232	10.0233	-0.064	0.0006%
$(3^1S)\ \eta_b(3S)$	10.3996	N/A	N/A	N/A
$(3^3S)\ \Upsilon(3S)$	10.4058	N/A	N/A	N/A
$(1^1P)\ h_b$	9.9002	9.8993	+0.945	0.009%
$(1^3P_0)\ \chi_{b0}$	9.8557	9.8594	-3.7	0.038%
$(1^3P_1)\ \chi_{b1}$	9.8986	9.8928	+5.8	0.059%
$(1^3P_2)\ \chi_{b2}$	9.9116	9.9122	-0.6	0.006%
$(2^1P)\ h_b(2P)$	10.2635	N/A	N/A	N/A
$(2^3P_0)\ \chi_{b0}(2P)$	10.2244	N/A	N/A	N/A
$(2^3P_1)\ \chi_{b1}(2P)$	10.2611	N/A	N/A	N/A
$(2^3P_2)\ \chi_{b2}(2P)$	10.2727	N/A	N/A	N/A
$(3^1P)\ h_b(3P)$	10.5579	N/A	N/A	N/A
$(3^3P_0)\ \chi_{b0}(3P)$	10.5215	N/A	N/A	N/A
$(3^3P_1)\ \chi_{b1}(3P)$	10.5555	N/A	N/A	N/A
$(3^3P_2)\ \chi_{b2}(3P)$	10.5666	N/A	N/A	N/A
$(1^1D)\ \eta_{b2}$	10.1490	N/A	N/A	N/A
$(1^3D_1)\ \Upsilon(1D)$	10.1415	N/A	N/A	N/A
$(1^3D_2)\ \Upsilon_2(1D)$	10.1497	N/A	N/A	N/A
$(1^3D_3)\ \Upsilon_3(1D)$	10.1518	N/A	N/A	N/A
$(2^1D)\ \eta_{b2}(2D)$	10.4540	N/A	N/A	N/A

State	Calculated (GeV)	Exp (PDG, GeV)	Abs Error (MeV)	% Error
(2^3D_1) $\Upsilon(2D)$	10.4463	N/A	N/A	N/A
(2^3D_2) $\Upsilon_2(2D)$	10.4544	N/A	N/A	N/A
(2^3D_3) $\Upsilon_3(2D)$	10.4570	N/A	N/A	N/A
(3^1D) $\eta_{b2}(3D)$	10.7195	N/A	N/A	N/A
(3^3D_1) $\Upsilon(3D)$	10.7117	N/A	N/A	N/A
(3^3D_2) $\Upsilon_2(3D)$	10.7198	N/A	N/A	N/A
(3^3D_3) $\Upsilon_3(3D)$	10.7227	N/A	N/A	N/A
Heavy-Light Meson ($b\bar{c}$)				
(1^1S) B_c	6.2756	6.2749	+0.781	0.012%
(1^3S) B_c^*	6.3152	6.3300	-14.71	0.232%
(2^1S) $B_c(2S)$	6.8787	6.8710	+7.712	0.112%
(2^3S) $B_c^*(2S)$	6.8921	6.8930	-0.885	0.012%
(3^1S) $B_c(3S)$	7.2654	N/A	N/A	N/A
(3^3S) $B_c^*(3S)$	7.2742	N/A	N/A	N/A
(1^1P) B_{c1}	6.7496	N/A	N/A	N/A
(1^3P_0) B_{c0}^*	6.6958	N/A	N/A	N/A
(1^3P_1) B_{c1}^*	6.7484	N/A	N/A	N/A
(1^3P_2) B_{c2}^*	6.7611	N/A	N/A	N/A
(2^1P) $B_{c1}(2P)$	7.1697	N/A	N/A	N/A
(2^3P_0) $B_{c0}^*(2P)$	7.1187	N/A	N/A	N/A
(2^3P_1) $B_{c1}^*(2P)$	7.1672	N/A	N/A	N/A
(2^3P_2) $B_{c2}^*(2P)$	7.1814	N/A	N/A	N/A
(3^1P) $B_{c1}(3P)$	7.5193	N/A	N/A	N/A
(3^3P_0) $B_{c0}^*(3P)$	7.4693	N/A	N/A	N/A
(3^3P_1) $B_{c1}^*(3P)$	7.5161	N/A	N/A	N/A
(3^3P_2) $B_{c2}^*(3P)$	7.5312	N/A	N/A	N/A
(1^1D) B_{c2}	7.0239	N/A	N/A	N/A
(1^3D_1) B_{c1}^*	7.0229	N/A	N/A	N/A
(1^3D_2) B_{c2}^*	7.0280	N/A	N/A	N/A
(1^3D_3) B_{c3}^*	7.0214	N/A	N/A	N/A
(2^1D) $B_{c2}(2D)$	7.3870	N/A	N/A	N/A
(2^3D_1) $B_{c1}^*(2D)$	7.3833	N/A	N/A	N/A
(2^3D_2) $B_{c2}^*(2D)$	7.3900	N/A	N/A	N/A
(2^3D_3) $B_{c3}^*(2D)$	7.3863	N/A	N/A	N/A
(3^1D) $B_{c2}(3D)$	7.7063	N/A	N/A	N/A
(3^3D_1) $B_{c1}^*(3D)$	7.7010	N/A	N/A	N/A
(3^3D_2) $B_{c2}^*(3D)$	7.7087	N/A	N/A	N/A
(3^3D_3) $B_{c3}^*(3D)$	7.7069	N/A	N/A	N/A

State	Calculated (GeV)	Exp (PDG, GeV)	Abs Error (MeV)	% Error

Table 2: Our errors findings for all 90 states.

Appendix C: Complete S-Wave GEM Expansion Coefficients (Charmonium Sector)

As an example of our found GEM parameters, it is beneficial to showcase one of the sectors. One can check below table for the 50 parameters for each excitatitons of the charmonium sector S-wave excitations.

Index	ν_{width} (GeV^2)	c_{1S} (Ground)	c_{2S} (1st Excited)	c_{3S} (2nd Excited)
0	400.0000	0.0001599	0.0001335	-0.0001388
1	257.2216	-0.0004651	-0.0003967	0.0004598
2	165.4074	0.0009751	0.0008519	-0.0010981
3	106.3659	-0.0013397	-0.0012299	0.0019027
4	68.3990	0.0018783	0.0018155	-0.0032698
5	43.9843	-0.0018758	-0.0020556	0.0048766
6	28.2843	0.0025495	0.0030033	-0.0080085
7	18.1883	-0.0018282	-0.0030378	0.0116355
8	11.6961	0.0033901	0.0053305	-0.0195036
9	7.5212	-0.0008874	-0.0048288	0.0281044
10	4.8365	0.0054909	0.0112117	-0.0485689
11	3.1102	0.0023539	-0.0084732	0.0688044
12	2.0000	0.0119963	0.0273415	-0.1236061
13	1.2861	0.0131690	-0.0143164	0.1681615
14	0.8270	0.0340093	0.0760029	-0.3251029
15	0.5318	0.0528735	-0.0069153	0.3905139
16	0.3420	0.1180149	0.2670420	-0.9627993
17	0.2199	0.2114630	0.2268324	0.5061600
18	0.1414	0.3750415	1.2964033	-4.3846484
19	0.0909	0.2406046	-0.8023682	9.0817596
20	0.0585	0.0022399	-1.4051017	-3.0422760
21	0.0376	0.0031003	0.2058551	-2.3964640
22	0.0242	-0.0014248	-0.0737477	0.6803200
23	0.0156	0.0004567	0.0215493	-0.1865907
24	0.0100	-0.0000783	-0.0035256	0.0295747

Appendix D: Code

All code is publicly shared in GitHub. Check LeventKaanOguz/400_2555365_lko